Falsifiable Hypothesis for a Federated Sovereign Genomics Cloud

A mid-term experimental framing for an OpenStack-based, data-local architecture in a resource-constrained setting

1. Research framing

Primary sovereignty target: operational sovereignty - the ability of each institution to continue local genomic ingest, storage, workflow execution and authorized access when the WAN becomes slow, intermittent or unavailable.

Secondary sovereignty targets: technical sovereignty (institution controls compute, storage and access services) and legal/data sovereignty (raw genomic data remains at the owning institution unless an explicitly authorized transfer is made).

Concept family: a deliberate hybrid of *decentralized/federated control* and *disconnection-tolerant computing*, implemented on institution-owned private-cloud resources. OpenStack is used as the infrastructure substrate; the federation exchanges metadata, policy decisions and approved analysis requests rather than centralizing raw genomes.

Proposed hypothesis (H1)

Under degraded or disconnected WAN conditions, a federated genomics architecture in which each institution keeps its own genomic object storage, local workflow execution and local identity enforcement will preserve local workflow availability and data residency better than a centralized architecture, while incurring only a measurable and bounded coordination overhead when connectivity is healthy.

Null hypothesis (H0)

There is no meaningful difference between the federated data-local architecture and the centralized baseline in local workflow availability, recovery behavior, data-residency violations, or performance degradation as WAN quality worsens.

Why this is falsifiable

The claim can fail. If local workflows stop when the WAN fails, if raw genomic objects cross the institutional boundary without authorization, if recovery is not faster than the centralized baseline, or if the healthy-network overhead becomes excessive, the hypothesis is not supported.

2. Experimental variables and measurable outcomes

Element	Proposed definition
Independent variable	WAN condition between institutional sites: normal, added latency, packet loss, bandwidth cap, and full partition.
Primary dependent variable	Local workflow availability / job success rate during WAN degradation and partition.
Secondary dependent variables	p50/p95/p99 latency, throughput, recovery time, data-loss window, CPU/memory footprint, bytes of raw genomic data crossing the WAN, and metadata reconciliation time after reconnect.
Controlled factors	Same VM sizes, same dataset, same workflow image, same storage policy, same number of clients, same run duration and same test sequence.
Operating context	Ugandan / resource-constrained institutional environment with intermittent WAN as the main stress condition.

Suggested acceptance criteria for the headline experiment

• **Data locality:** 0 unauthorized bytes of FASTQ/BAM/CRAM/VCF leave the owning institution during all trials.

• **Partition survival:** workflows whose required inputs are already local continue to start and complete during a WAN partition.

• **Recovery:** after reconnect, metadata/policy state reconciles without raw-data re-upload and without manual reconstruction of completed local jobs.

• **Performance:** healthy-link federation overhead is reported, not hidden; p95/p99 latency and throughput are compared against the mandatory baselines.

• **Statistics:** each configuration is repeated at least 30 times and reported with mean plus 95% confidence interval.

Test data recommendation

Use public or synthetic genomic test data rather than identifiable patient data. A small FASTQ-to-VCF workflow is sufficient for the systems experiment; the research question is about architecture behavior, not biological discovery.

3. Comparison plan aligned to the mid-term

Architecture arm	What it represents	What you measure
Your federated design	Independent OpenStack site(s) + local object storage + local workflow execution + metadata/policy federation	Availability under partition, data locality, reconciliation, latency, throughput, recovery
OpenStack private-cloud baseline	Centralized on-premises sovereign cloud	Same metrics under identical load and network degradation
Lightweight Kubernetes baseline	Constrained-node / edge baseline (k3s/MicroK8s/k0s)	Resource footprint and degraded-network behavior
Centralized single-node baseline	Naive floor	Cost of distribution and minimum achievable latency
CRANE Cloud baseline	Managed-PaaS contrast arm	Operational comparison from available deployment/documented baseline

One-sentence contribution claim to test - not to assume

A data-local federated genomics cloud can trade a modest coordination overhead during normal connectivity for substantially better continuity and sovereignty when inter-site connectivity degrades or disappears.

What would count as a successful mid-term demonstration?

Run the same small genomic workflow on the federated design and a centralized OpenStack baseline; inject latency/loss/bandwidth limits; then cut the inter-site link. Show that locally available workflows continue, raw genomic objects remain local, metrics are captured, and federation state reconciles after the link returns. Report both successes and failures.

Source basis: MCS 7101 mid-term brief, especially the requirements for one falsifiable hypothesis, fixed baselines, repeated measurements, Rust-based instrumentation, DDIL testing, and production validation.